

Problem Set #8 — Delay Coordinate Embedding

University of Colorado - Boulder
CSCI4446 — Chaotic Dynamics

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1 Data Sampling Results

Figure 1 shows the state space with the sampling rate set to every hundredth data entry. The velocity values were constructed using first order forward divided differences. The points appear to spiral inwardly, which makes sense as the data represent the position of a damped pendulum. With a damped pendulum, we would expect it to oscillate with smaller and smaller magnitude as it approached the steady state (hanging straight down). The choice of sampling rate was somewhat arbitrary. I attempted to produce this behavior with other values, but $\frac{1}{100}$ produced the “cleanest” spiral, indicating that the pendulum was sampled about 100 times as fast as the sensor could reliably capture changes in the data. The imperfections in the spiral can be attributed to inaccuracies in the sensor.

2 Embedding

2.1 Embedding data2

Figure 2 shows a state space reconstruction of the data from **data2** where the time series data were reconstructed into a seven-dimensional reconstruction space ($m = 7$). The time interval for embedding was set to $\tau = 0.15$ seconds, and the axes represent the 0th and 2nd entries in the reconstructed state space vector. That is, $\theta(t)$ is represented by the x -coordinate and $\theta(t + 2\tau) = \theta(t + 0.3)$ seconds is represented by the y -coordinate. The data were sampled at every fifth entry ($dt = 0.01$) of the set to avoid oversampling. The result is a chaotic attractor, meaning the driven pendulum is running at a frequency such that it has nonrepeating and highly unpredictable behavior.

2.2 Embedding data3

The higher-frequency driven pendulum sampled in **data3**, when formed in reconstructed state space, forms a quasiperiodic attractor, indicated by the toroidal, ribbon-like band through the middle of the plot. The pendulum never repeats exactly, but it remains bound to a specific region of state

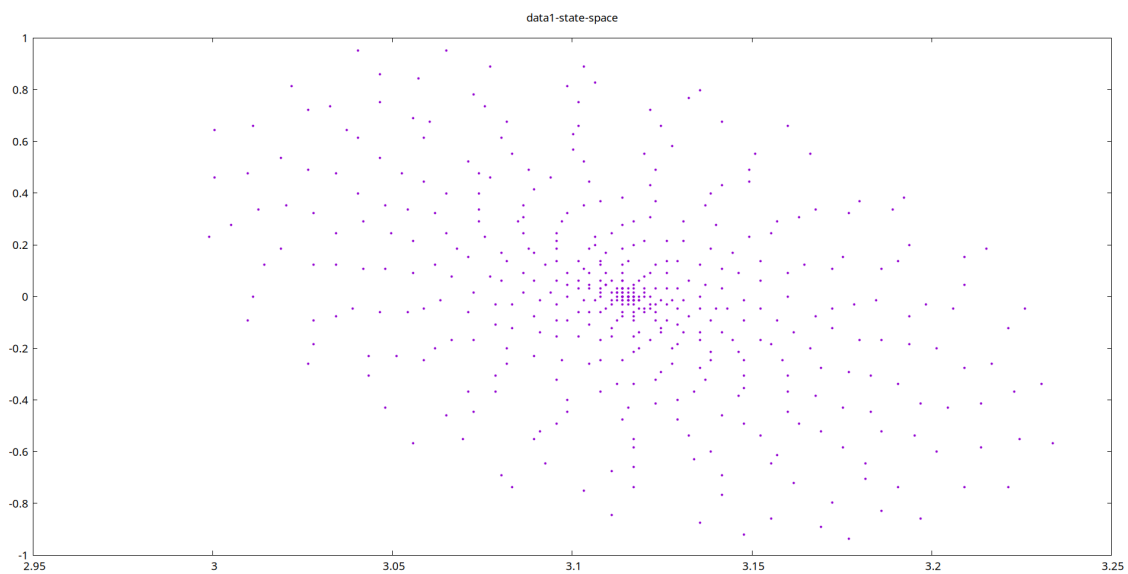


Figure 1: State Space Reconstructed from `data1`

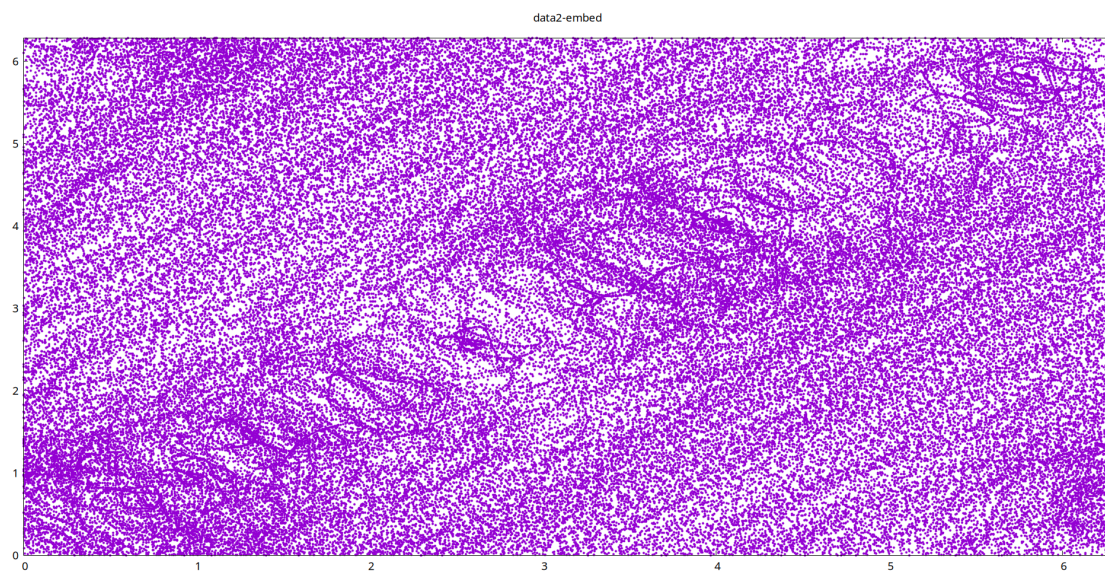


Figure 2: Delay Coordinate Embedding on `data2`
with $\tau = 0.15\text{s}$

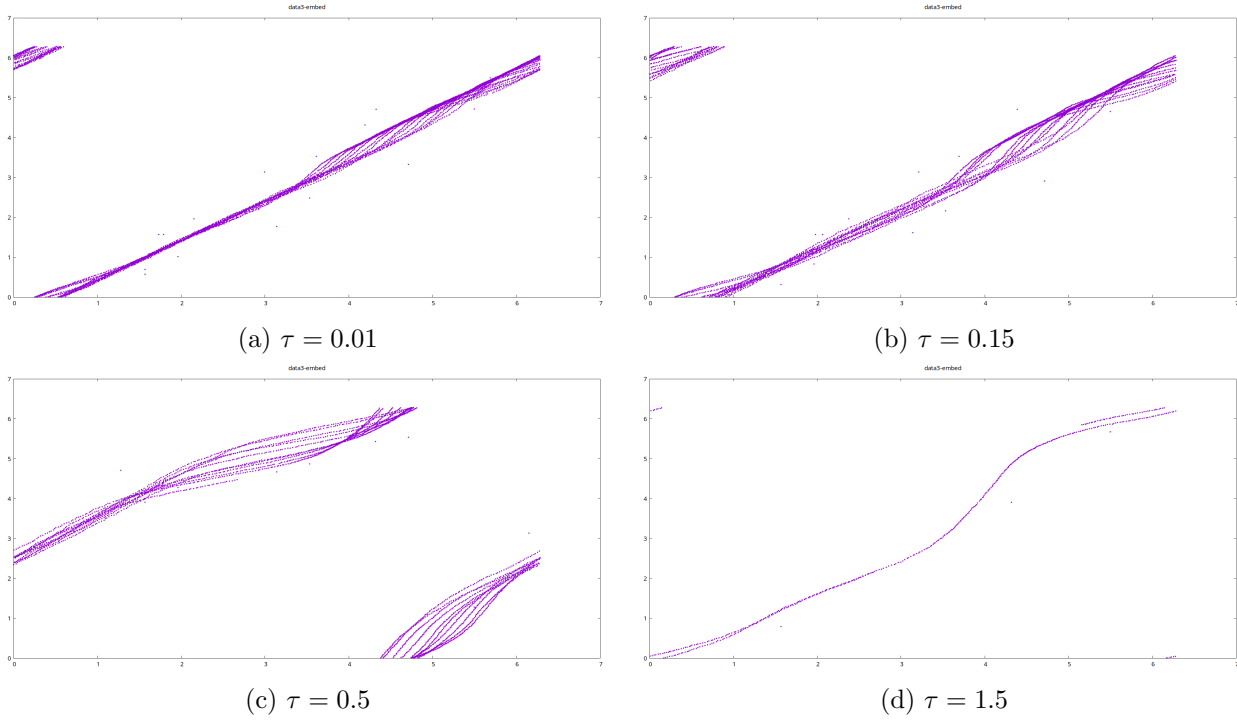


Figure 3: Delay Coordinate Embeddings of `data3`

space. The four plots in Figure 3 represent different embeddings of the data, indicated by the value of τ in the caption. For large and small τ , the data are being sampled too closely or too far apart to see the behavior, but for the right values, we can see the torus fold over itself several times during the loop, demonstrating its quasiperiodic structure.

3 Thought Experiments

3.1 Requirements from Takens' Theorem

The undriven pendulum is a two-dimensional system, containing the position of the pendulum and its rate of change. The drive phase adds a third dimension to the system in the driven case. Takens' Theorem requires $m > 2d$ dimensions in reconstruction space to capture the dynamics, meaning m must be set to at least 5 in the undriven case and 7 in the driven case.

3.2 Alternative m ?

If m were set to 2, the embedding would not be able to capture the dynamics correctly, resulting in “crossings” in reconstruction space that do not exist in actual state space. If m were set to 25, it would represent the data in an unrealistically-high dimensional space, amplifying noise.

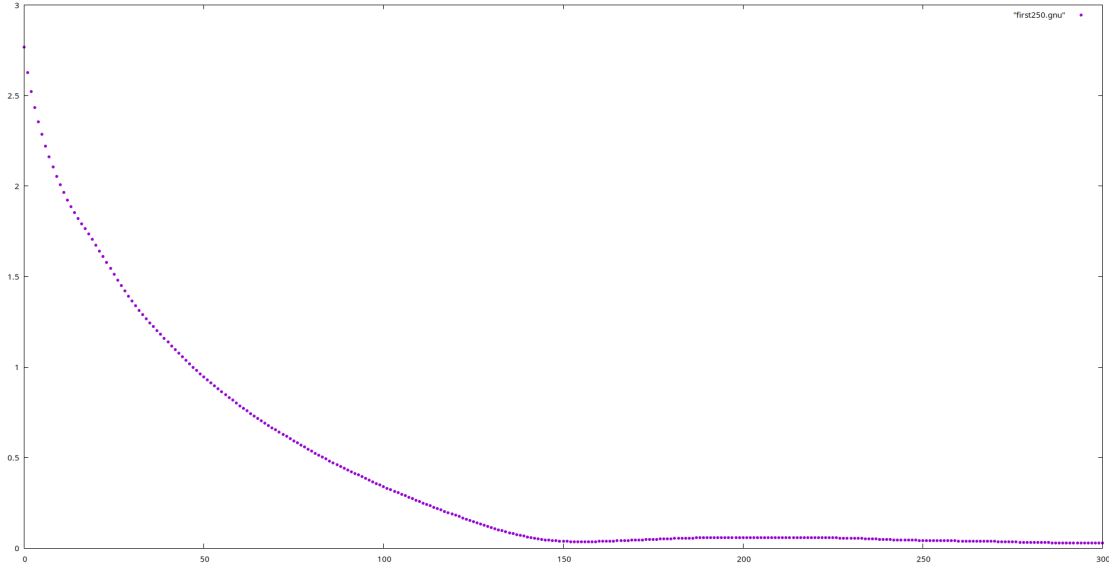


Figure 4: Average Mutual Information for the First 300 Time Steps

3.3 High and Low τ

For extremely small τ , the data would be too highly correlated to examine the dynamics, forming a diagonal line in reconstruction. For extremely large τ , the data would be too unrelated to establish any patterns—the x -coordinate would be mapped to a y -coordinate that represents the position of the pendulum at a completely irrelevant point in time.

4 First Minimum of Mutual Information

Using the `mutual` program from TISEAN, I plotted the average mutual information between $\theta(t)$ and $\theta(t + \tau)$ for the first 300 values of τ , as shown in Figure 4. From the data, I determined the first minimum occurred when τ was set to 155 time steps, i.e. $\tau = 0.31$ seconds. The specific TISEAN call I used was

```
mutual -D 300 ./input/data2.first250sec -o ./output/first250.gnu
```

5 False Near Neighbors

After finding the First Minimum, I took that value and used it to find an appropriate embedding dimension using TISEAN's `false_nearest` tool. I ran

```
false_nearest -M 1,10 -d 155 ./input/data2.first250sec -o
./output/false_nearest.gnu
```

and produced the plot in Figure 5. The value of m at which the ratio of false near neighbors dropped below ten percent was $m = 8$.

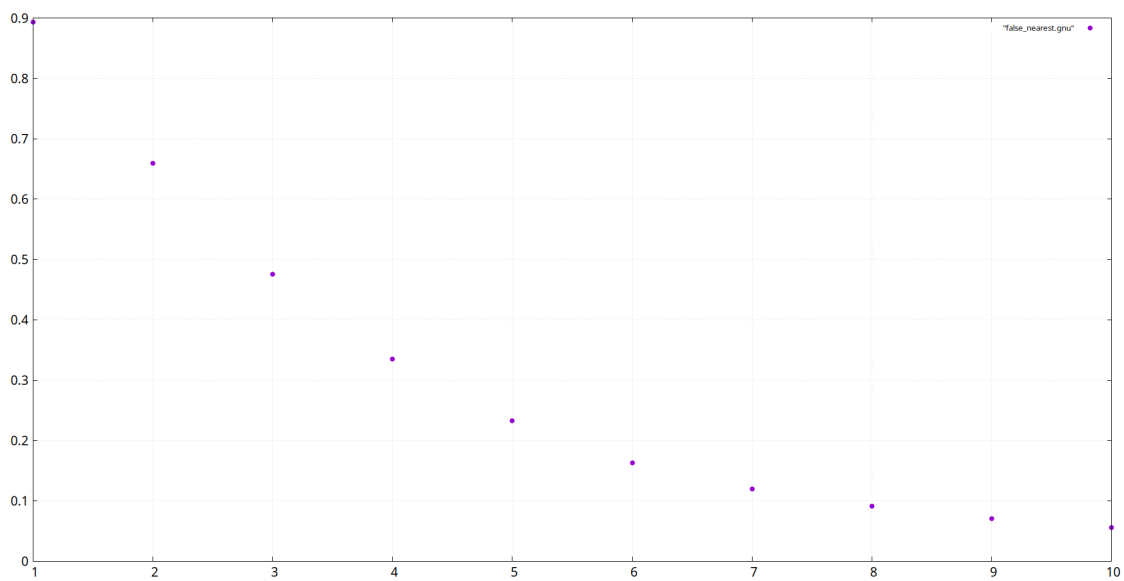


Figure 5: Ratio of False Near Neighbors for Different Embedding Dimensions